

# User Evaluation of Dynamic X-Ray Vision in Mixed Reality

Hung-Jui Guo<sup>✉\*</sup>

The University of Texas at Dallas

Jonathan Z. Bakdash<sup>†</sup>Laura R. Marusich<sup>‡</sup>

U.S. DEVCOM Army Research Laboratory

Omeed Eshaghi Ashtiani<sup>§</sup>Balakrishnan Prabhakaran<sup>||</sup>

The University of Texas at Dallas

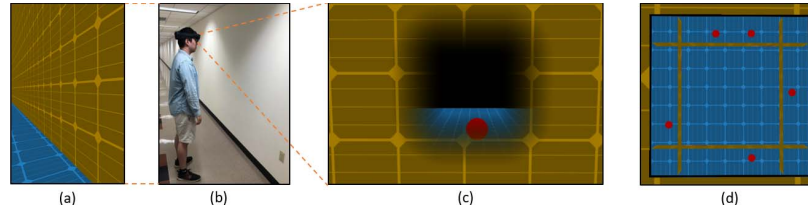


Figure 1: (a) Observed and created virtual wall in the generated augmented reality environment based on an indoor room, (b) Physical wall in the same indoor room with a HoloLens 2 user (c) Dynamic X-Ray vision window to see through virtual wall to find a red target object (d) Minimap view, red circles indicate the target objects.

## ABSTRACT

X-Ray vision is a technique providing a Superman-like ability to help users see through physical obstacles. In this paper, we utilized the HoloLens 2 [3] to build a virtual environment and a dynamic X-Ray vision window based on the created environment. Unlike previous works, our system allows users to physically move and still have the X-Ray vision ability in real-time, providing depth cues and additional information in the surrounding environment. Additionally, we designed an experiment and recruited multiple participants as a proof of concept for our system's ability.

**Index Terms:** Human-centered computing—Mixed / augmented reality; Human-centered computing—Collaborative interaction

## 1 INTRODUCTION

X-Ray vision is an ability used by Superman and other fictional superheroes to see visual information occluded by solid physical objects. Research in Augmented Reality (AR) has tried to make X-Ray vision feasible by utilizing the real-virtual interaction characteristic of AR to help users gain information about otherwise occluded areas, while still maintaining the context of the occluded object and beyond-line-of-sight environment with direct visual information.

Existing X-Ray vision applications allow users to gain information about occludee behind occluder but only from a fixed position. In contrast, we proposed a dynamic X-Ray vision system that updates with users' physical movements [2]. In this system virtual content is rendered dynamically in the Microsoft HoloLens 2 device [3], so the surrounding virtual environment and virtual objects update in real-time according to users' changing position, movement, and eye-gaze direction in the physical environment. This system simulates the appearance of a physical window (as shown in Figure 1 (c)), and provides visual depth cues typically available in real-world environments [1] when using the X-Ray vision ability.

\*This paper is an extended version of our demo paper entitled "Dynamic X-Ray Vision in Mixed Reality" published in VRST 2022 [2].

Corresponding author: Hung-Jui Guo, e-mail: hxg190003@utdallas.edu

<sup>†</sup>e-mail: jonathan.z.bakdash.civ@army.mil

<sup>‡</sup>e-mail: laura.m.cooper20.civ@army.mil

<sup>§</sup>e-mail: omeed.ashtiani@utdallas.edu

<sup>||</sup>e-mail: bprabhakaran@utdallas.edu

Here, we focus on evaluating the effectiveness of our X-Ray vision system for users. We conducted a behavioral experiment comparing our X-Ray vision system with a minimap, an alternative technique commonly used for spatial navigation in video-games.

## 2 DYNAMIC X-RAY VISION SYSTEM DESIGN

Dynamic X-Ray vision can dynamically move and adapt based on user's movement. Our system combined scene understanding, eye tracking, and clipping primitive to implement the X-Ray vision feature. Scene understanding is a pre-built feature in the Mixed Reality Toolkit 2 (MRTK 2), which uses the long-throw depth camera to capture user's surrounding physical environment to build a virtual environment, as shown in Figure 2.

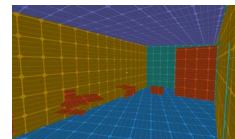


Figure 2: Generated AR environment based on an indoor room.

To achieve X-Ray vision in this environment, we used a clipping primitive feature, a box-like object that can make specified contact areas partially transparent to help users see through the occluder to the occludee. The size of the window can be changed by the users to match the size they want and how much occluder information to keep. Furthermore, we also included the eye-tracking feature in HoloLens 2 to make the X-Ray vision automatically follow user's eye movement in real-time. Therefore, by including the above features, our dynamic X-Ray vision system can dynamically update based on users' position/movement as they physically navigating the surrounding physical environment. This X-Ray vision has the potential to provide benefits on better depth cues over prior implementations. Therefore, we ran a behavioral experiment to evaluate the user effectiveness of our system.

## 3 EVALUATION EXPERIMENTAL DESIGN

To evaluate how our system helps users perceive information behind the occluder and provide real-time depth cues [4, 5] with physical movement, we build a **target tagging game** as proof of concept.

### 3.1 Target Tagging Game Design

For experimental control and consistency across conditions, we built a square virtual room of size 4 meters x 4 meters. Five virtual target

objects (red spheres with a diameter of 0.3 meters) were placed 0.5 meters behind the occluder walls, preventing direct visibility. For each participant, the position of each target was randomly chosen between 0.3 and 3.7 meters in the horizontal dimension and between 0.3 and 1.9 meters in the vertical dimension of its occluding wall. Additionally, the target placement was constrained so that the space beyond each of the four walls contained at least one target.

The participants' goal was to find and tag the target objects behind the occluders. Participants in the experimental condition received the X-Ray vision window capability, in which the window size was fixed to 0.6 meters x 0.6 meters. Those in the control condition received a fixed orientation minimap capability, which showed the top view of the created virtual room, as shown in Figure 1 (d).

### 3.2 Participants and Procedure

We recruited 11 university students (9 male and 2 female) to participate in this experiment. Participants ranged in height from 1.62 meters to 1.93 meters, and in age from 25 to 35. Participants performed the target tagging task in both experimental conditions with the order of each condition randomly counterbalanced.

During the experiment process, we recorded the start time for each actual trial and the target objects' tag time. After the experiment, participants completed a post-questionnaire, assessing gender, age, height in meters, Head-Mounted Display (HMD) experience, and time spend playing videogames per week. These questions can help us analyze different potential factors that could be associated with experiment completion time.

## 4 EXPERIMENTAL RESULTS AND DISCUSSION

We first examined the total time the participants spent to complete tagging 5 target objects using different abilities.

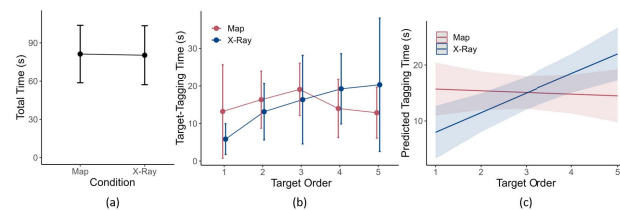


Figure 3: All error bars are 95% onfidence intervals. (a) Average total time to tag five targets for each experimental condition. (b) Average tagging times for targets by sequential order. (c) Visualization of predicted interaction effect by experimental condition and target order.

As shown in Figure 3 (a), there was no statistically significant difference in total time for the two experimental conditions using paired-samples  $t$ -test,  $t(10) = 0.12$ ,  $p = 0.90$ . However, the minimap in this experiment provides the complete view of the surrounding environment and the position of the target objects, while the X-Ray vision condition requires participants to physically walk to find and view the target objects. Therefore, having a similar average total time with the minimap ability suggests effectiveness of this system that includes the physical movement characteristic.

Figure 3 (b) shows the tagging-time for each target in order. A multi-level model assessing the effect of both target order and experimental condition on participants' target-tagging times indicates a significant interaction effect  $p = 0.005$ , visualized in Figure 3 (c). In the X-Ray condition, participants appear to be faster to tag the first target object, with longer times for subsequent targets, while tagging times are relatively stable across the minimap condition. It may be the case that users are initially focused on the entire scene when they have access to a minimap, but that users with X-Ray vision capability can quickly focus on tagging their first target.

Figure 4 (a) shows the relationship between time spent on video games per week and target tagging time, which was not statistically

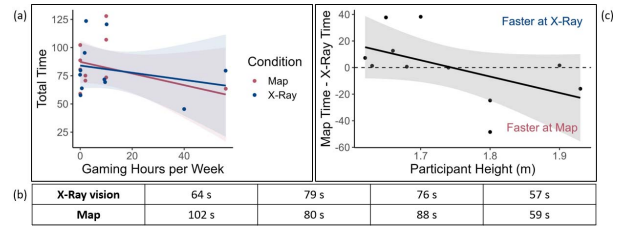


Figure 4: (a) Relationship between gaming time and total time in the experimental task (b) Target tagging time for the four participants with 0 hours of gaming per week. (c) The relationship between participants' height and the total tagging time in two different abilities.

significant. However, as shown in Figure 4 (b), all four participants with 0 hours of gaming per week showed roughly equal or faster times in X-Ray vision compared to the minimap condition. This suggests that the dynamic X-Ray vision system may be fairly user-friendly for novices.

We also assessed the effect of participants' heights on total tagging time in the two conditions. Figure 4 (c) shows the relative advantage of the X-Ray condition (total time in X-Ray minus total time in minimap) as a function of participant height. This pattern suggests that taller participants may perform better using the minimap, whereas shorter participants perform better with X-Ray vision.

We showed dynamic X-ray vision has similar average effectiveness to a minimap in a target-tagging task. However, we found evidence of trade-offs between the two conditions, including differing effects of target order and participant height. Future directions should include further investigation of these trade-offs with larger sample sizes, using a larger or user-adjustable X-Ray window, a combination of X-Ray vision and the minimap, and/or a minimap using relative orientation (track-up).

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### REFERENCES

- [1] J. E. Cutting and P. M. Vishton. Perceiving layout and knowing distances: The integration, relative potency, and contextual use of different information about depth. In *Perception of space and motion*, pp. 69–117. Elsevier, 1995.
- [2] H.-J. Guo, J. Bakdash, L. Marusich, and B. Prabhakaran. Dynamic x-ray vision in mixed reality. In *Proceedings of the 28th ACM Symposium on Virtual Reality Software and Technology*, pp. 1–2, 2022. doi: 10.1145/3562939.3565675
- [3] Microsoft. Hololens, 2015. <https://www.microsoft.com/en-us/hololens>.
- [4] N. Phillips, F. A. Khan, B. Kruse, C. Bethel, and J. E. Swan. An x-ray vision system for situation awareness in action space. In *2021 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 593–594, 2021. doi: 10.1109/VRW52623.2021.00179
- [5] S. Zollmann, R. Grasset, G. Reitmayr, and T. Langlotz. Image-based x-ray visualization techniques for spatial understanding in outdoor augmented reality. In *Proceedings of the 26th Australian Computer-Human Interaction Conference on Designing Futures: The Future of Design, OzCHI '14*, p. 194–203. Association for Computing Machinery, New York, NY, USA, 2014. doi: 10.1145/2686612.2686642